University (Tenant)

User (Abstract)

- Admin
- Faculty
- Student

Course

Lecture

Assignment

Quiz/Test

Submission

University Onbording
Admin registers university
→ university_id created
→ super admin created

Admin Flow
Admin → creates Course
Admin → assigns Faculty
Admin → enrolls Students
Assign faculty to courses

Faculty Flow
Faculty → Course
→ Lecture upload
→ Assignment create
→ Quiz create
Evaluate submissions

Student Flow
Enroll in courses
Watch lectures
Submit assignments
Attempt quizzes
View grades

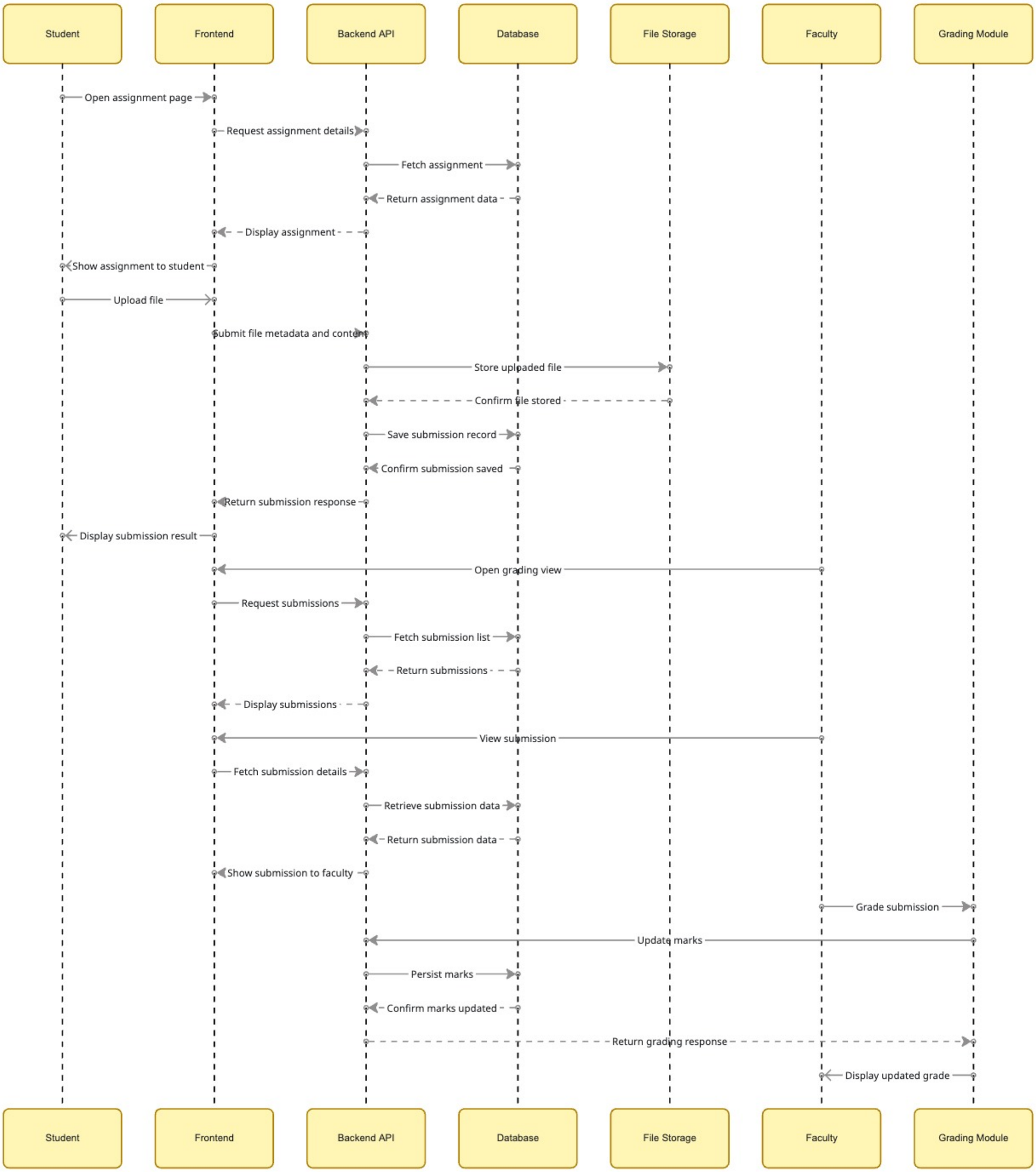
Course
├── Lectures
├── Assignments
├── Quizzes
├── Enrollments (Students)
└── Faculty

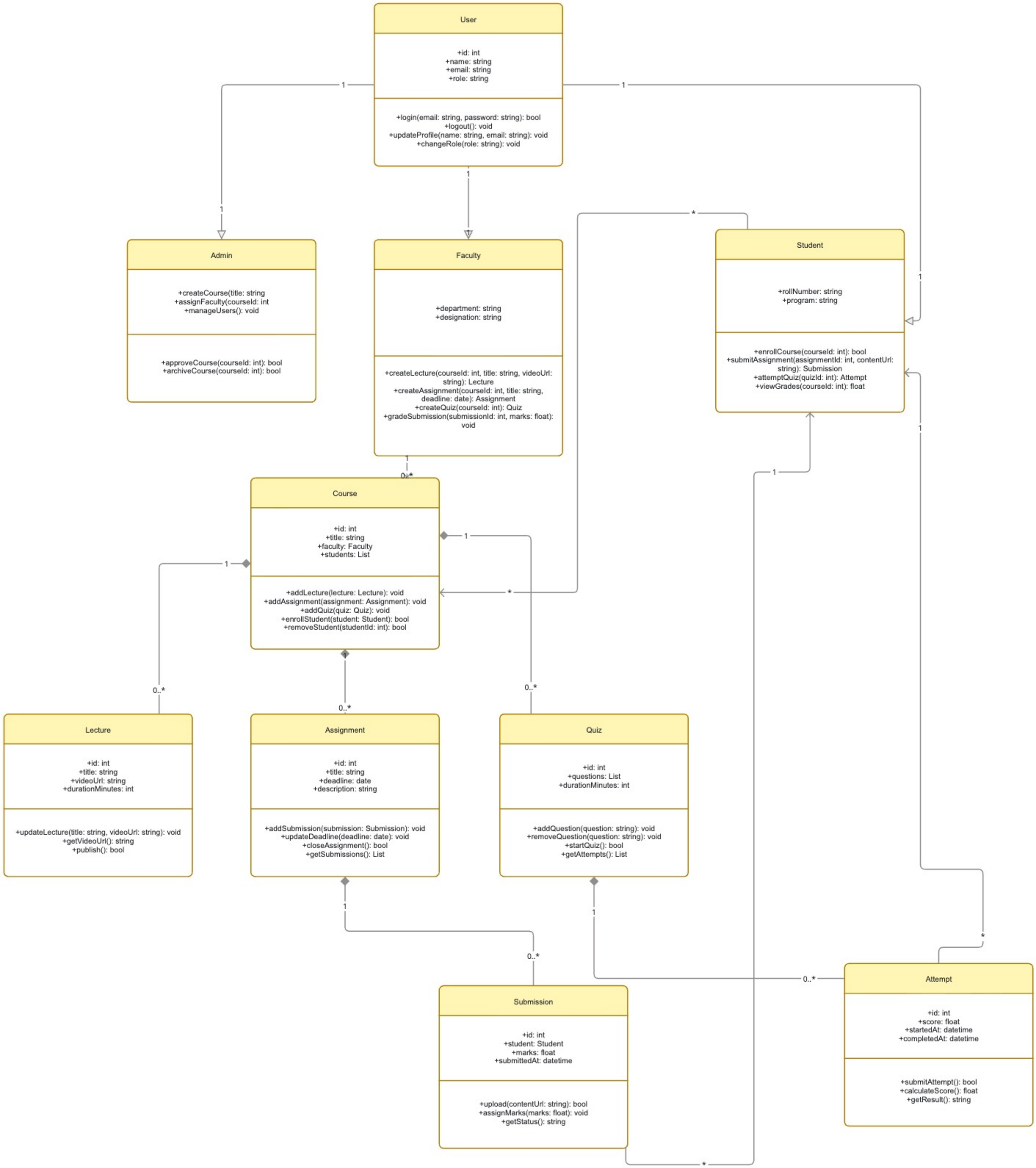
Example Flow
Student opens dashboard
→ sees enrolled courses

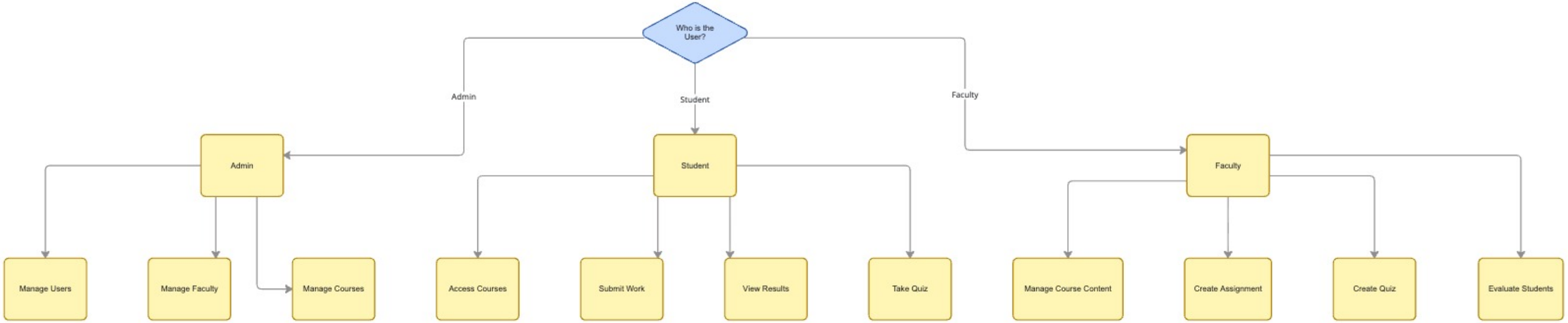
Clicks course
→ sees lectures + assignments + quizzes

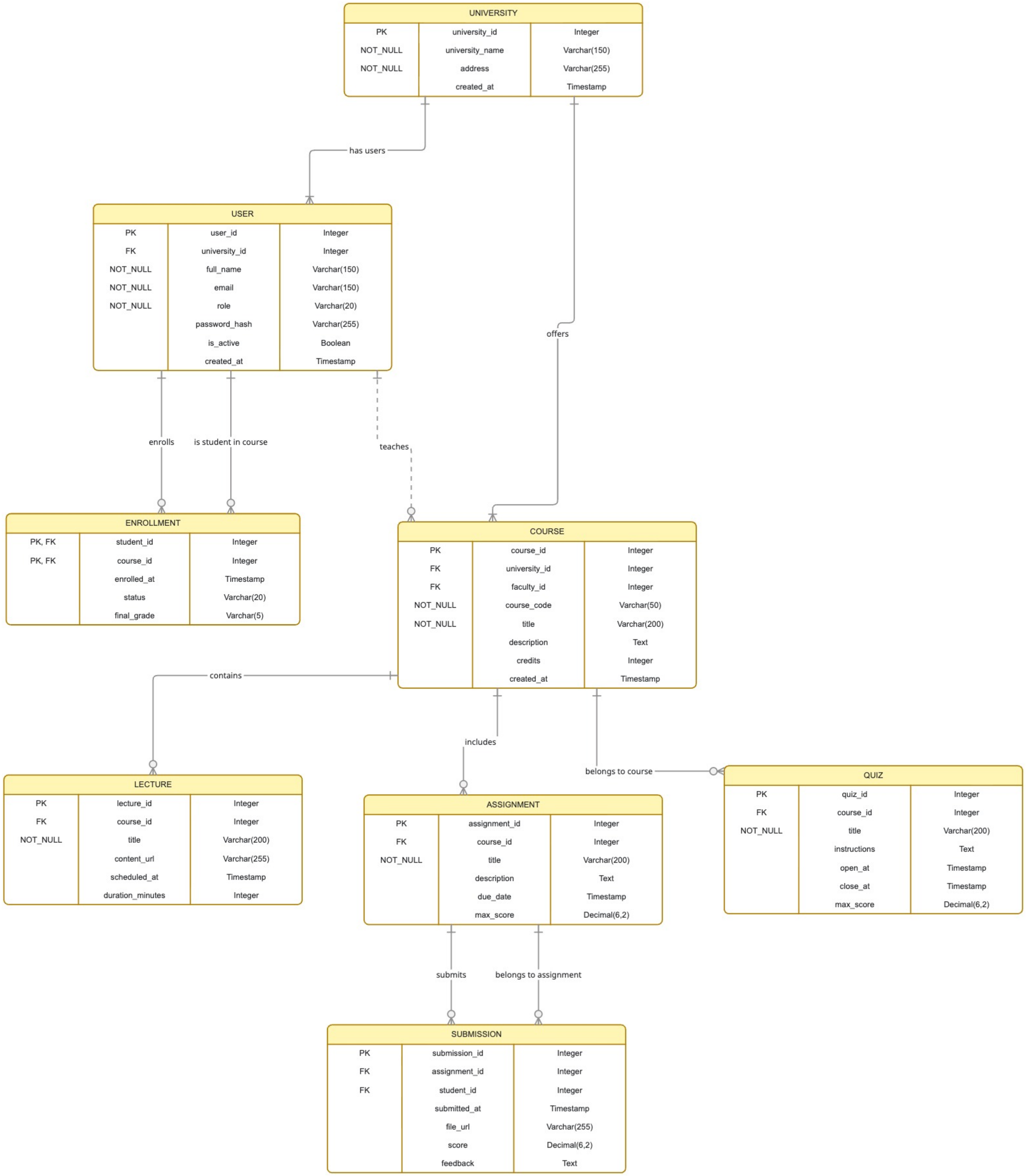
Submits assignment
→ stored in submissions table

Faculty checks submission
→ gives marks


Student
Frontend
Backend API
Database
File Storage
Faculty
Grading Module
Open assignment page
Request assignment details
Fetch assignment
Return assignment data
Display assignment
Show assignment to student
Upload file
Submit file metadata and content
Store uploaded file
Confirm file stored
Save submission record
Confirm submission saved
Return submission response
Display submission result
Open grading view
Request submissions
Fetch submission list
Return submissions
Display submissions
View submission
Fetch submission details
Retrieve submission data
Return submission data
Show submission to faculty
Grade submission
Update marks
Persist marks
Confirm marks updated
Return grading response
Display updated grade
Student
Frontend
Backend API
Database
File Storage
Faculty
Grading Module


User
+id: int
+name: string
+email: string
+role: string
+login(email: string, password: string): bool
+logout(): void
+updateProfile(name: string, email: string): void
+changeRole(role: string): void
Admin
+createCourse(title: string
+assignFaculty(courseId: int
+manageUsers(): void
+approveCourse(courseId: int): bool
+archiveCourse(courseId: int): bool
Faculty
+department: string
+designation: string
+createLecture(courseId: int, title: string, videoUrl: string): Lecture
+createAssignment(courseId: int, title: string, deadline: date): Assignment
+createQuiz(courseId: int): Quiz
+gradeSubmission(submissionId: int, marks: float): void
Student
+rollNumber: string
+program: string
+enrollCourse(courseId: int): bool
+submitAssignment(assignmentId: int, contentUrl: string): Submission
+attemptQuiz(quizId: int): Attempt
+viewGrades(courseId: int): float
Course
+id: int
+title: string
+faculty: Faculty
+students: List
+addLecture(lecture: Lecture): void
+addAssignment(assignment: Assignment): void
+addQuiz(quiz: Quiz): void
+enrollStudent(student: Student): bool
+removeStudent(studentId: int): bool
Lecture
+id: int
+title: string
+videoUrl: string
+durationMinutes: int
+updateLecture(title: string, videoUrl: string): void
+getVideoUrl(): string
+publish(): bool
Assignment
+id: int
+title: string
+deadline: date
+description: string
+addSubmission(submission: Submission): void
+updateDeadline(deadline: date): void
+closeAssignment(): bool
+getSubmissions(): List
Quiz
+id: int
+questions: List
+durationMinutes: int
+addQuestion(question: string): void
+removeQuestion(question: string): void
+startQuiz(): bool
+getAttempts(): List
Submission
+id: int
+student: Student
+marks: float
+submittedAt: datetime
+upload(contentUrl: string): bool
+assignMarks(marks: float): void
+getStatus(): string
Attempt
+id: int
+score: float
+startedAt: datetime
+completedAt: datetime
+submitAttempt(): bool
+calculateScore(): float
+getResult(): string
1
0..*
*


Who is the User?
Admin
Student
Faculty
Admin
Student
Faculty
Manage Users
Manage Faculty
Manage Courses
Access Courses
Submit Work
View Results
Take Quiz
Manage Course Content
Create Assignment
Create Quiz
Evaluate Students


UNIVERSITY
PK university_id Integer
NOT_NULL university_name Varchar(150)
NOT_NULL address Varchar(255)
created_at Timestamp
has users
USER
PK user_id Integer
FK university_id Integer
NOT_NULL full_name Varchar(150)
NOT_NULL email Varchar(150)
NOT_NULL role Varchar(20)
password_hash Varchar(255)
is_active Boolean
created_at Timestamp
offers
enrolls
is student in course
teaches
ENROLLMENT
PK, FK student_id Integer
PK, FK course_id Integer
enrolled_at Timestamp
status Varchar(20)
final_grade Varchar(5)
COURSE
PK course_id Integer
FK university_id Integer
FK faculty_id Integer
NOT_NULL course_code Varchar(50)
NOT_NULL title Varchar(200)
description Text
credits Integer
created_at Timestamp
contains
includes
belongs to course
LECTURE
PK lecture_id Integer
FK course_id Integer
NOT_NULL title Varchar(200)
content_url Varchar(255)
scheduled_at Timestamp
duration_minutes Integer
ASSIGNMENT
PK assignment_id Integer
FK course_id Integer
NOT_NULL title Varchar(200)
description Text
due_date Timestamp
max_score Decimal(6,2)
QUIZ
PK quiz_id Integer
FK course_id Integer
NOT_NULL title Varchar(200)
instructions Text
open_at Timestamp
close_at Timestamp
max_score Decimal(6,2)
submits
belongs to assignment
SUBMISSION
PK submission_id Integer
FK assignment_id Integer
FK student_id Integer
submitted_at Timestamp
file_url Varchar(255)
score Decimal(6,2)
feedback Text